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A spectral theory for turbulent burning velocity across scales

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Damköhler's two hypotheses attribute the enhancement of the turbulent burning velocity s_T over the laminar value s_L to an increase in flame-surface area by eddies larger than the flame thickness δ_L, and to an increase in effective diffusivity by smaller eddies. In intense turbulence, the measured s_T/s_L exceeds the measured area ratio several-fold, an excess that Nivarti *et al.* (*J. Fluid Mech.*, vol. 858, 2019, R1) attributed to a sub-δ_L diffusivity integrated over a model energy spectrum. Here that integral is extended to the whole spectrum and made dimensionally consistent by taking the diffusivity at each wavenumber from the spectral eddy diffusivity of two-point closures. For a model spectrum, the total enhancement reduces to $C\,\mathcal{F}(Re)\,u'/s_L$, where u' is the turbulence intensity, C is a constant and $\mathcal{F}$ is a weak, saturating function of the turbulence Reynolds number Re. The resulting relation $s_T/s_L = [1 + C\,\mathcal{F}(Re)\,u'/s_L]^{1/2}$ recovers Damköhler's linear growth of s_T with u' in weak turbulence and the sublinear 'bending' in strong turbulence, and its sub-δ_L part recovers the $Ka^{1/3}$ scaling of Nivarti *et al.*, where Ka is the Karlovitz number. The relation has the same form as the power-law flame-wrinkling factor of Charlette *et al.* (*Combust. Flame*, vol. 131, 2002, p. 159) with exponent $1/2$, which supports the square-root dependence at large scales. With spectral constants from non-reacting turbulence, the prediction follows 80 measurements from eight datasets spanning four decades of Ka and Re, 71 % of them within a factor of 1.5.

Key words: combustion, turbulent reacting flows, turbulent mixing

1. Introduction

Damköhler (1940) proposed two mechanisms by which turbulence raises the burning velocity of a premixed flame above its laminar value s_L. Eddies larger than the laminar flame thickness δ_L wrinkle the flame without altering its inner structure, so that the turbulent burning velocity

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s_T grows with the flame-surface area,

$$\frac{s_T}{s_L} = \frac{A_T}{A} = 1 + \frac{\Delta A}{A}, \tag{1.1}$$

where A_T is the wrinkled area and A the mean (unwrinkled) area. Eddies smaller than δ_L instead enhance transport within the flame as a diffusive process. Since $s_L \propto D^{1/2}$ for a fixed reaction time (Zel'dovich & Frank-Kamenetskiĭ 1938), an effective turbulent diffusivity $D_T = D + \Delta D$ yields

$$\frac{s_T}{s_L} = \left(\frac{D_T}{D}\right)^{1/2} = \left(1 + \frac{\Delta D}{D}\right)^{1/2}. \tag{1.2}$$

The two hypotheses, (1.1) and (1.2), are conventionally assigned to separate regimes, delineated by the Karlovitz number $Ka \equiv \delta_L^2/\eta^2$, where η is the Kolmogorov scale (Peters 1999, 2000). Throughout, unity Schmidt and Lewis numbers are assumed, so that $\nu = D = s_L\delta_L$, and the Reynolds and Karlovitz numbers are

$$Re \equiv \frac{u'\ell_0}{\nu} = \frac{u'}{s_L}\frac{\ell_0}{\delta_L}, \quad Ka = \left(\frac{u'}{s_L}\right)^{3/2}\left(\frac{\ell_0}{\delta_L}\right)^{-1/2}, \quad \text{such that} \quad \frac{u'}{s_L} = Ka^{1/2}Re^{1/4}, \tag{1.3}$$

with u' the r.m.s. velocity fluctuation and ℓ_0 the integral length scale.

Planar measurements in intense turbulence (Yuen & Gülder 2013; Wabel *et al.* 2017) show that (1.1) fails for $Ka > 1$: the area ratio A_T/A saturates near 4–6 while s_T/s_L continues to rise, reaching nearly 30 (figure 1*b*; see also the reviews of Driscoll *et al.* 2020 and Hochgreb 2023). Direct numerical simulations at $u'/s_L = 35$ show a smaller excess, about 14 % on average, attributed to flame collisions (Poludnenko & Oran 2011; Hicks 2015). Nivarti *et al.* (2019) reconciled s_T and A_T/A by writing $s_T/s_L = (A_T/A)(1 + \Delta D^{>}/D)^{1/2}$ and estimating the sub-δ_L enhancement $\Delta D^{>}$ by summing Prandtl mixing-length diffusivities over the portion of a model energy spectrum with wavenumbers $\kappa > \delta_L^{-1}$. The result, $\Delta D^{>}/D \sim Ka^{2/3}$, accounted for the measured ratio $(s_T/s_L)/(A_T/A) \sim Ka^{1/3}$. Related spectral estimates of small-scale enhancement have been made by Zimont (1979), Ronney & Yakhot (1992) and Gülder (2007), and spectral closures have previously been used for the large-scale regime (Peters 1992; Chaudhuri *et al.* 2011, 2012).

Because area enhancement by large eddies is itself a function of the spectrum, it is natural to ask whether a single spectral sum over all scales can account for s_T in every regime (Walker 2019). Two difficulties must be overcome. First, the scale velocity $u(\kappa) = [2E(\kappa)]^{1/2}$ used by Nivarti *et al.* (2019), where $E(\kappa)$ is the energy spectrum function, has dimensions of $\mathrm{L}^{3/2}\,\mathrm{T}^{-1}$, so the integrand $u(\kappa)/\kappa$ is not a diffusivity and its prefactor is arbitrary. Second, the square root in (1.2) derives from laminar-flame theory, and there is no obvious reason for it to hold at scales where area enhancement contributes significantly to the burning velocity.

This paper resolves both. In § 2 the scale-dependent diffusivity is taken from the spectral eddy diffusivity of two-point closures (Kraichnan 1976; Chollet & Lesieur 1981; Chollet 1984), which fixes its dimensions and prefactor. In § 3 the resulting integral is shown to reduce to a function of u'/s_L and Re alone, and the reduced relation is shown to coincide with the power-law flame-wrinkling factor of Charlette *et al.* (2002*a*) with exponent $1/2$. Section 4 compares the prediction with eight experimental datasets.

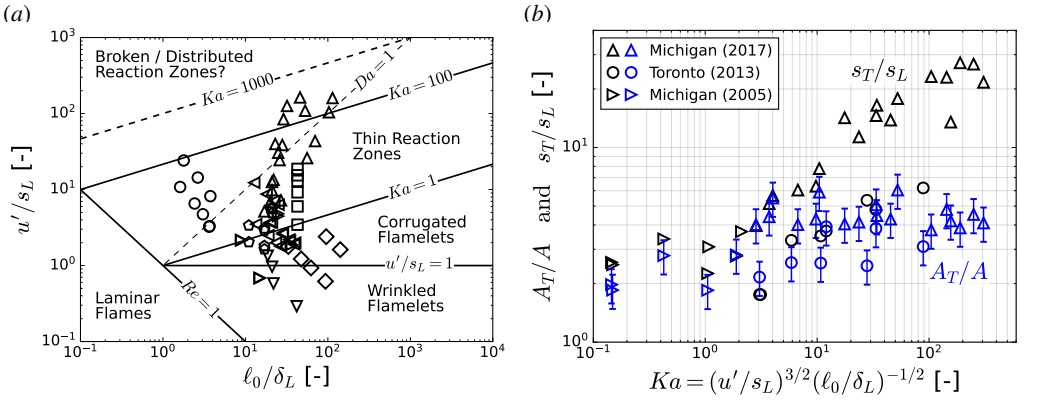


Figure 1: (*a*) Experimental conditions on the turbulent combustion regime diagram. (*b*) Burning-velocity ratio s_T/s_L (black) and flame-surface-area ratio A_T/A (blue, with ±20 % uncertainty from Wabel 2017; Yuen 2009; Filatyev *et al.* 2005) against Ka for the datasets where both were measured; for Filatyev *et al.* (2005), A_T/A is the squared ratio of wrinkled to mean flame perimeter, multiplied by a $2/\pi$ consistency factor for comparing slot and round burners. Symbols and data sources are listed in table 1.

2. Formulation

2.1. *Spectral eddy diffusivity*

In two-point closures of isotropic turbulence, the depletion of energy from a mode κ to all modes beyond a cut-off $\kappa_c > \kappa$ is represented by a spectral eddy viscosity (Kraichnan 1976; Chollet & Lesieur 1981; Pope 2000),

$$\nu_e(\kappa|\kappa_c) = \nu_e^+(\kappa/\kappa_c)\left[\frac{E(\kappa_c)}{\kappa_c}\right]^{1/2}, \tag{2.1}$$

where the non-dimensional function ν_e^+, computed with the eddy-damped quasi-normal Markovian closure, is flat for $\kappa \ll \kappa_c$ and rises to a cusp as $\kappa \to \kappa_c$ (Chollet 1984). Chollet (1984) also computed the corresponding eddy diffusivity D_e^+ of a passive scalar, whose ratio to ν_e^+ defines a spectral turbulent Schmidt number $Sc_T = \nu_e^+/D_e^+$. Fits to these curves (figure 2*b*; coefficients in supplementary table S1) give, at the cusp, $\nu_e^+(1) = 0.868$, $Sc_T(1) = 0.462$ and hence $D_e^+(1) = 1.88$.

The factor $[E(\kappa)/\kappa]^{1/2}$ in (2.1) can be written u_κ/κ, where $u_\kappa \equiv [\kappa E(\kappa)]^{1/2}$ is the characteristic velocity of eddies of size κ^{-1}. It is therefore Prandtl's mixing-length product of a velocity and a length at that scale, with the dimensions of a diffusivity. Accordingly, we assign to each wavenumber the scalar eddy diffusivity at the cusp,

$$D_\kappa(\kappa) \equiv D_e^+(1)\left[\frac{E(\kappa)}{\kappa}\right]^{1/2} = D_e^+(1)\,\frac{u_\kappa}{\kappa}, \tag{2.2}$$

that is, the transport exerted by eddies of wavenumber κ on scales just larger than themselves. Equation (2.2) replaces the integrand $[2E(\kappa)]^{1/2}/\kappa$ of Nivarti *et al.* (2019), removing the dimensional inconsistency and taking the prefactor from the closure rather than setting it to unity.

2.2. *Total diffusivity enhancement*

Following Nivarti *et al.* (2019), the contributions of all scales are summed with wavenumber measured in units of the flame wavenumber δ_L^{-1}. Using $D = s_L\delta_L$,

$$\frac{\Delta D}{D} = \frac{1}{D}\int_{\kappa_0}^{\kappa_\eta} D_\kappa \,\mathrm{d}(\kappa\delta_L) = \frac{D_e^+(1)}{s_L}\int_{\kappa_0}^{\kappa_\eta} u_\kappa \,\mathrm{d}\ln\kappa, \tag{2.3}$$

with $\kappa_0 = \ell_0^{-1}$ and $\kappa_\eta = \eta^{-1}$. This weighting is the central modelling assumption of the theory: each logarithmic band of eddies adds a diffusivity $D_e^+ u_\kappa \delta_L$, with the flame thickness as its mixing length. It differs from the classical estimate $\Delta D \propto u'\ell_0$, in which every eddy mixes over its own size (Damköhler 1940). The two can be compared through the flux each band carries across the flame. Let $\Delta\Lambda$ be the change in a progress variable Λ across the flame, so that the molecular flux is $D\Delta\Lambda/\delta_L = s_L\Delta\Lambda$. A band of velocity u_κ can transport at most $u_\kappa\Delta\Lambda$. With mixing length κ^{-1}, its gradient-diffusion flux would be $u_\kappa(\kappa\delta_L)^{-1}\Delta\Lambda$. For eddies smaller than the flame ($\kappa\delta_L > 1$) this is below that bound, and replacing κ^{-1} by δ_L changes only an $O(1)$ constant, because both sums are dominated by $\kappa \approx \delta_L^{-1}$. For eddies larger than the flame the gradient-diffusion flux exceeds the bound, so their mixing length is capped at δ_L. Each band then carries a flux of order $u_\kappa\Delta\Lambda$, which relative to the molecular flux gives (2.3). The estimate $\Delta D \propto u'\ell_0$ is therefore valid only in Damköhler's limit $\ell_0 < \delta_L$. A further consequence of (2.3) is that δ_L is absent from the enhancement factor, which becomes a ratio of velocities.

For the energy spectrum, we adopt the model of Pope (2000) used by Nivarti *et al.* (2019),

$$E(\kappa) = C\varepsilon^{2/3}\kappa^{-5/3} f_L(\kappa\ell_0) f_\eta(\kappa\eta), \quad f_L = \left[1 + \frac{c_L}{(\kappa\ell_0)^2}\right]^{-(5/3+p_0)/2}, \quad f_\eta = \mathrm{e}^{-\beta\kappa\eta}, \tag{2.4}$$

where ε is the kinetic energy dissipation rate. In units of the laminar flame, with $\hat\kappa \equiv \kappa\delta_L$, the dissipation rate $\varepsilon = 15\nu(u'/\lambda)^2$ with Taylor microscale $\lambda = \ell_0(10/Re)^{1/2}$ (Taylor 1935) becomes $\hat\varepsilon \equiv \varepsilon\delta_L/s_L^3 = (3/2)Ka^2$ (Nivarti *et al.* 2019). The integration limits become $\hat\kappa_0 = Ka^{1/2}Re^{-3/4}$ and $\hat\kappa_\eta = Ka^{1/2}$ (Walker 2019). Substituting (2.4) into (2.3) gives

$$\mathcal{S} \equiv \frac{\Delta D}{D} = C\,Ka^{2/3}\int_{Ka^{1/2}Re^{-3/4}}^{Ka^{1/2}} \hat\kappa^{-4/3}\,\hat f_L^{1/2}\hat f_\eta^{1/2}\,\mathrm{d}\hat\kappa, \qquad C \equiv D_e^+(1)\,C^{1/2}(3/2)^{1/3}, \tag{2.5}$$

where $\hat f_L = [1 + c_L KaRe^{-3/2}\hat\kappa^{-2}]^{-(5/3+p_0)/2}$ and $\hat f_\eta = \exp(-\beta\hat\kappa Ka^{-1/2})$. Compared with Nivarti *et al.* (2019, their equation (2.7)), the exponent of the integrand changes from $-11/6$ to $-4/3$, and the prefactor $(2C)^{1/2}(3/2)^{1/3}$ is replaced by C, which equals 2.64 for $C = 1.5$. Finally, (1.2) is applied over the entire spectrum,

$$\frac{s_T}{s_L} = (1 + \mathcal{S})^{1/2}, \tag{2.6}$$

a step justified in § 3.2.

3. Analysis

3.1. *Reduction to two groups*

Equation (2.5) simplifies considerably. Substituting $\hat\kappa = \hat\kappa_0\kappa^*$, where $\kappa^* \equiv \kappa\ell_0$, and using $Ka^{1/2}Re^{1/4} = u'/s_L$ from (1.3), the prefactor $Ka^{2/3}\hat\kappa_0^{-1/3}$ reduces to u'/s_L and the integration

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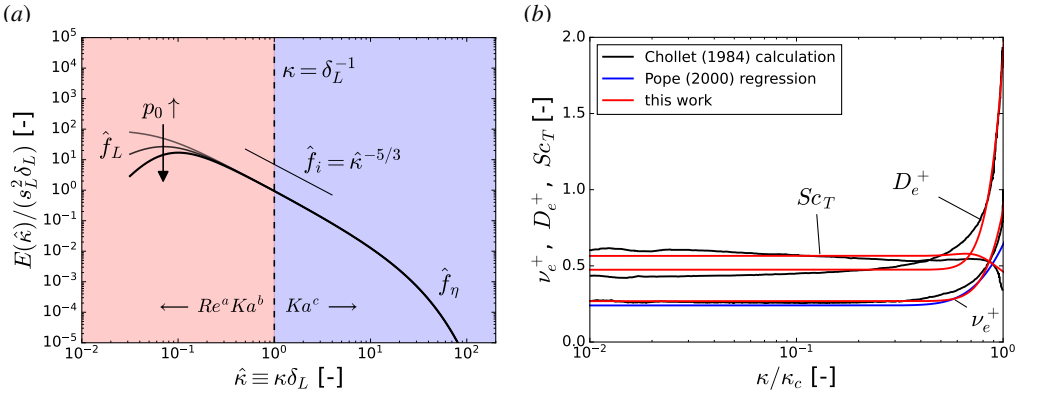


Figure 2: (*a*) Model energy spectrum (2.4) in flame units for increasing p_0. Wavenumbers below δ_L^{-1} (red) contribute a mixed *Re*–*Ka* dependence to (2.5), and those above δ_L^{-1} (blue) a dependence on *Ka* alone. (*b*) Spectral eddy viscosity ν_e^+, eddy diffusivity D_e^+ and turbulent Schmidt number Sc_T computed by Chollet (1984), the regression for ν_e^+ given by Pope (2000), and those used here.

limits become 1 and $Re^{3/4}$. The enhancement is therefore

$$\mathcal{S} = C\,\frac{u'}{s_L}\,\mathcal{F}(Re), \qquad \mathcal{F}(Re) = \int_1^{Re^{3/4}} \kappa^{*-4/3}\left(1+\frac{c_L}{\kappa^{*2}}\right)^{-(5/3+p_0)/4} \exp\left(-\frac{\beta\kappa^*}{2Re^{3/4}}\right) \mathrm{d}\kappa^*, \tag{3.1}$$

and (2.6) becomes

$$\frac{s_T}{s_L} = \left[1 + C\,\mathcal{F}(Re)\,\frac{u'}{s_L}\right]^{1/2}. \tag{3.2}$$

This is equivalent to $\Delta D = C\,\mathcal{F}(Re)\,u'\delta_L$, an eddy diffusivity with the r.m.s. velocity as its velocity scale and the flame thickness as its mixing length. The burning velocity thus depends on two groups only, u'/s_L and *Re*; the Karlovitz number is not an independent parameter.

The function $\mathcal{F}$ carries the spectral contribution. With $f_L = f_\eta = 1$ (inertial range only), $\mathcal{F} = 3(1 - Re^{-1/4})$. Retaining the dissipation range ($f_L = 1$), $\mathcal{F}$ can be written in terms of the upper incomplete gamma function $\Gamma(s,x)$,

$$\begin{aligned}\mathcal{F} &= (\beta/2)^{1/3} Re^{-1/4}\left[\Gamma(-1/3,\,\beta Re^{-3/4}/2) - \Gamma(-1/3,\,\beta/2)\right] \\ &= 3 - c_\beta Re^{-1/4} + O(Re^{-3/4}),\end{aligned} \tag{3.3}$$

with $c_\beta = (\beta/2)^{1/3}[\Gamma(-1/3,\,\beta/2) - \Gamma(-1/3)] = 5.61$ for $\beta = 5.2$. The energy-containing range reduces the asymptote to $\mathcal{F}_\infty = \int_1^\infty \kappa^{*-4/3}(1 + c_L\kappa^{*-2})^{-(5/3+p_0)/4}\,\mathrm{d}\kappa^*$, which equals 2.56, 2.24 and 2.05 for $p_0 = 0$, 2 and 4 ($c_L = 6.78$). For $p_0 = 0$, $\mathcal{F} = 0.98$, 1.60 and 2.00 at $Re = 10^2$, 10^3 and 10^4. Since $\mathcal{F}$ approaches its asymptote as $Re^{-1/4}$, the Reynolds-number dependence persists up to the highest laboratory Reynolds numbers, but it is weak (figure 3*a*). The dissipation range lowers $\mathcal{F}$ more than the energy-containing range for $Re \lesssim 10^3$, and less for larger *Re*.

Three limits of (3.2) connect it with earlier results.

(i) *Weak turbulence*, $C\,\mathcal{F}\,u'/s_L \ll 1$: $s_T/s_L \approx 1 + \frac{1}{2}C\,\mathcal{F}\,u'/s_L$. This is Damköhler's linear scaling $s_T - s_L \propto u'$, which he derived from (1.1) by taking the area increase proportional to u'/s_L.

(ii) *Strong turbulence*, $C\,\mathcal{F}\,u'/s_L \gg 1$: $s_T/s_L \approx (C\,\mathcal{F}\,u'/s_L)^{1/2}$, i.e. $s_T/s_L \propto Ka^{1/4}Re^{1/8}$

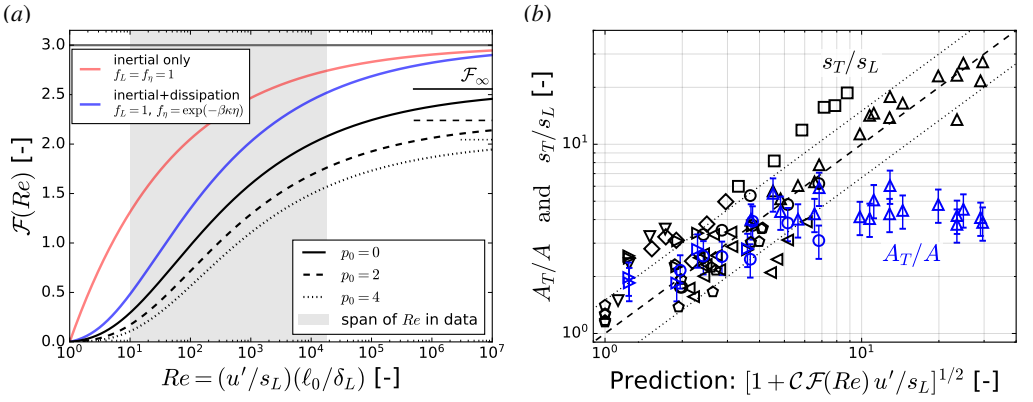


Figure 3: (*a*) The function $\mathcal{F}(Re)$ of (3.1) for $p_0 = 0$, 2 and 4, with the inertial-range result $3(1 - Re^{-1/4})$ and the form (3.3) with $f_L = 1$; short lines at the right mark $\mathcal{F}_\infty$, and shading spans the Re of the measurements. (*b*) Measured s_T/s_L (black) and A_T/A (blue) against the prediction (3.2). Dashed line, equality; dotted lines, a factor of 1.5 either side. Symbols as in table 1.

at fixed $\mathcal{F}$. The transition between (i) and (ii) is the 'bending' of s_T with increasing u' (Bradley 1992; Lipatnikov & Chomiak 2002). It arises here without any quenching mechanism, from the square root in (2.6) alone. For comparison, taking $\Delta D \propto u'\ell_0$ in (1.2) gives $s_T/s_L \propto Re^{1/2}$ (Damköhler 1940; Chaudhuri *et al.* 2011); Chaudhuri *et al.* (2012) found the same scaling in expanding flames with the mean flame radius in place of ℓ_0. The difference lies in the mixing length: ℓ_0 in that estimate, δ_L in (2.3).

(iii) *Sub-flame scales.* The part of (2.5) with $\hat{\kappa} > 1$, for which $\hat{f}_L \approx 1$ when $\ell_0 \gg \delta_L$, is

$$\begin{aligned} \mathcal{S}^> &= C(\beta/2)^{1/3} Ka^{1/2} \left[\Gamma(-1/3,\, \beta Ka^{-1/2}/2) - \Gamma(-1/3,\, \beta/2)\right] \\ &= 3CKa^{2/3} - c_\beta C Ka^{1/2} + O(Ka^{1/6}), \end{aligned} \tag{3.4}$$

which retains the leading-order $Ka^{2/3}$ scaling of Nivarti *et al.* (2019) and hence their $(s_T/s_L)/(A_T/A) \sim Ka^{1/3}$, now with a dimensionally consistent integrand. In the inertial range, the remaining large-scale part is $\mathcal{S}^< \approx 3C(u'/s_L - Ka^{2/3})$, so that $\mathcal{S}^</\mathcal{S}^> \approx (\ell_0/\delta_L)^{1/3} - 1$. Eddies larger than the flame therefore dominate once $\ell_0/\delta_L \gtrsim 8$. The total, however, is a smooth function of u'/s_L and Re in which δ_L plays no distinct role, so a division of turbulence into 'large' and 'small' scales at δ_L is not needed to predict s_T.

3.2. *Equivalence with power-law flame wrinkling*

Charlette *et al.* (2002*a*) modelled the unresolved wrinkling of a flame in a large-eddy simulation with filter width Δ as a power law between an outer cut-off Δ and an inner cut-off η_c,

$$\Xi_\Delta = \left(1 + \frac{\Delta}{\eta_c}\right)^{\beta_w}, \qquad \frac{\Delta}{\eta_c} = \Gamma_\Delta \frac{u'_\Delta}{s_L} \quad \longrightarrow \quad \Xi_\Delta = \left(1 + \Gamma_\Delta \frac{u'_\Delta}{s_L}\right)^{\beta_w}. \tag{3.5}$$

Here η_c follows from an equilibrium between the production of surface by sub-filter strain, $\Gamma_\Delta u'_\Delta/\Delta$, and its destruction by propagation, s_L/η_c. The efficiency function Γ_Δ is a spectral superposition of the straining by all eddies smaller than Δ (Meneveau & Poinsot 1991; Colin *et al.* 2000). Evaluated at $\Delta = \ell_0$, the limit noted by Charlette *et al.* (2002*a*), where $\Xi_{\ell_0} = s_T/s_L$, and with $\beta_w = 1/2$, (3.5) is identical to (3.2) with $\Gamma_{\ell_0} \equiv C\,\mathcal{F}(Re)$.

Equation (3.2) can therefore be read in two ways: as Damköhler's second hypothesis

	Dataset	Configuration	ϕ [-]	u'/s_L [-]	ℓ_0/δ_L [-]	$\overline{\Lambda}$ [-]	n
△	Michigan (Wabel *et al.* 2017)	piloted Bunsen, 21.6 mm	0.75	5.2–163	17–114	0.5	18
▷	Michigan (Filatyev *et al.* 2005)	piloted slot, 25.4 mm	1.0	0.68–4.8	8.9–37	0.5	7
○	Toronto (Yuen & Gülder 2013)	piloted Bunsen, 11.2 mm	0.6–1.0	3.2–24	1.6–3.8	0.5	8
◁	Toronto (Tamadonfar & G. 2014)	piloted Bunsen, 11.1 mm	0.7–1.0	1.8–12	13–40	0.5	16
▽	Tohoku (Kobayashi *et al.* 1996)	piloted Bunsen, 20 mm	0.9	0.29–1.4	19–42	0.5	4
□	Aachen (Plessing *et al.* 2000)	low-swirl, 50.8 mm	0.7	3.5–19	43	0.5	6
◇	Ottawa (Smallwood *et al.* 1995)	Bunsen, 25 mm	0.7–1.0	0.62–2.4	33–141	0.05	6
⬠	Damköhler (1940)	Bunsen, 1.4–2.7 mm	–	0–5.8	11–21	–	15

Table 1: Datasets, with the symbols used in figures 1, 3 and 4. All are methane–air flames at 0.1 MPa and about 300 K except those of Damköhler (1940) (propane–butane–oxygen). Here ϕ is the equivalence ratio, $\overline{\Lambda}$ the mean progress variable of the isosurface defining s_T, and n the number of points.

with a spectral eddy diffusivity, or as his first hypothesis with a power-law wrinkling of exponent $1/2$ and a spectrally derived efficiency. The burning velocity cannot distinguish between them. Charlette *et al.* (2002*a*) chose $\beta_w = 0.5$ from comparisons with measured s_T, and their dynamic procedure (Charlette *et al.* 2002*b*) returned local exponents concentrated between 0.5 and 0.6 (supplementary figure S1). They also argued that the fractal value $\beta_w = 1/3$, required to recover Damköhler's large-u' limit $s_T \propto u'$, is unlikely to be reached in practice. The square root in (2.6) is thus consistent with the observed behaviour of large-scale wrinkling, which resolves the second difficulty raised in § 1. The equivalence concerns the consumption rate rather than the area itself. The sub-δ_L contribution (3.4) acts through transport within the flame (Nivarti *et al.* 2019) and is not expected to appear in a measured A_T/A.

4. Comparison with experiments

Table 1 lists the experiments, which span $0.02 < Ka < 310$ and $10 < Re < 2{\times}10^4$ (figure 1*a*). For uniformity, s_T is the global consumption speed on the $\overline{\Lambda} = 0.5$ isosurface wherever the source reports it (supplementary section S2), and Re and Ka are evaluated from the tabulated u'/s_L and ℓ_0/δ_L using (1.3). The spectral constants are those of non-reacting turbulence: $C = 1.5$, $\beta = 5.2$ and $c_L = 6.78$ (Pope 2000), with $D_e^+(1) = 1.88$ from Chollet (1984), so that $C = 2.64$. We take $p_0 = 0$, a spectrum that is flat at low wavenumbers, which gave the best agreement in the comparisons of Walker (2019); $p_0 = 2$ and 4 lower the predictions by medians of 12 % and 20 % over the present data. Apart from this choice, no constant is adjusted to the data.

Figure 3(*b*) compares (3.2) with the measurements. Over a more than twenty-fold range of s_T/s_L, 71 % of the 80 points lie within a factor of 1.5 of the prediction, and the r.m.s. logarithmic deviation is 0.38. The two high-Ka datasets are captured best. For Michigan (Wabel *et al.* 2017), at $3 < Ka < 310$ and Re up to 2×10^4, the geometric-mean ratio of measurement to prediction is 1.06 with an r.m.s. logarithmic deviation of 0.21; Toronto (Yuen & Gülder 2013) has the same mean ratio with a deviation of 0.20. These are the conditions under which A_T/A (blue symbols) falls furthest below s_T/s_L, and the prediction tracks s_T rather than the area. The remaining datasets show systematic offsets: the low-Ka Bunsen flames of Tohoku, Ottawa and Michigan (2005) lie above the prediction by factors of 1.4–1.8, the low-swirl flames of Aachen by a factor of 2, and the Toronto data of Tamadonfar

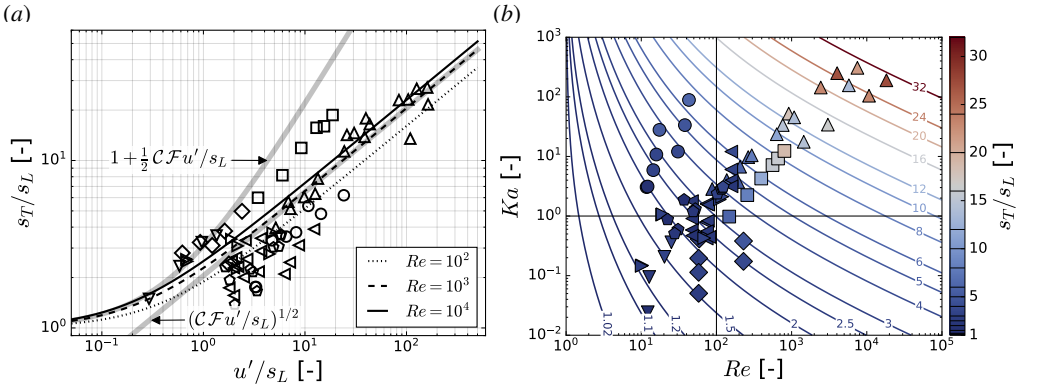


Figure 4: (*a*) Measured s_T/s_L against u'/s_L, with (3.2) at three Reynolds numbers; thick grey lines show the weak- and strong-turbulence limits at $Re = 10^3$. (*b*) Contours of (3.2) in the (Re, Ka) plane, labelled with the predicted s_T/s_L; symbols are coloured by the measured s_T/s_L on the same scale. Symbols as in table 1.

& Gülder (2014) at 0.8 times the prediction. Offsets of this size are comparable to the effect of the isosurface choice alone, which changes the reported s_T by up to a factor of two in the Toronto data (Yuen & Gülder 2013). They are also comparable to the known sensitivity of s_T to flame configuration (Driscoll 2008; Verma & Lipatnikov 2016). There is, however, a weak trend with Ka: points with $Ka \leqslant 1$ lie above the prediction by a factor of 1.3 on average, whereas those with $Ka > 1$ are unbiased (mean ratios of 1.00 for $1 < Ka \leqslant 10$ and 1.07 for $Ka > 10$). One candidate for the low-Ka excess is hydrodynamic (Darrieus–Landau) instability, which adds wrinkling when u'/s_L is small (Chaudhuri *et al.* 2011) and is absent from the present formulation.

Figure 4(*a*) shows the data against u'/s_L. The prediction passes from the linear growth of limit (i) to the square-root growth of limit (ii) near $u'/s_L \approx 1/(\mathcal{C}\mathcal{F})$, which is 0.2–0.4 for $10^2 \leqslant Re \leqslant 10^4$. The data follow this bending over three decades of u'/s_L, and the spread in $\mathcal{F}(Re)$ between $Re = 10^2$ and 10^4 accounts for part of their scatter at fixed u'/s_L. Figure 4(*b*) maps the prediction over the regime landscape. At $Re \gtrsim 10^3$, where $\mathcal{F}$ is nearly constant, the contours approach lines of constant u'/s_L, $Ka \propto Re^{-1/2}$. At lower Re, the decrease of $\mathcal{F}$ steepens them, and there s_T becomes sensitive to Re at fixed u'/s_L. The measured values (symbol colour) increase across the plane in the same direction as the contours, from the low-Re, low-u'/s_L corner to the high-Re, high-Ka Michigan flames. Supplementary figure S3 shows the same measurements as $(s_T/s_L)/Ka$ against Re.

5. Conclusions

The diffusive enhancement of the burning velocity proposed by Nivarti *et al.* (2019) for sub-flame eddies has been extended to the whole turbulence spectrum and made dimensionally consistent. Its scale-dependent diffusivity is taken from the spectral eddy diffusivity of two-point closures. For a model spectrum, the total enhancement reduces to $\mathcal{C}\,\mathcal{F}(Re)\,u'/s_L$, with $\mathcal{F}$ a weak, saturating function of the Reynolds number. The resulting burning velocity, (3.2), recovers three established results: Damköhler's linear law in weak turbulence, the bending of s_T in strong turbulence, and the $Ka^{1/3}$ excess of s_T/s_L over A_T/A at high Karlovitz number. With spectral constants from non-reacting turbulence, it follows eight datasets over four decades of Ka and Re, mostly to within a factor of 1.5. The same expression is the power-law wrinkling factor of Charlette *et al.* (2002*a*) with exponent 1/2, so area enhancement by large

eddies and diffusive enhancement by small eddies need not be treated as distinct mechanisms when predicting s_T.

Two implications follow. First, evaluated at a filter scale Δ with the sub-filter intensity u'_Δ and Reynolds number Re_Δ, (3.2) provides a sub-filter wrinkling factor of the form (3.5) whose efficiency function, $C\mathcal{F}(Re_\Delta)$, follows from the spectrum rather than from a fit to combustion data. Second, the remaining differences between datasets may be explained by the quantities the theory neglects: anisotropy of the approaching flow and the effect of heat release on its spectrum, differences in the definitions of s_T and δ_L, and differential diffusion, stretch and quenching. Measurements of the turbulence spectrum upstream of the flame, and simulations at matching u'/s_L and Re in different configurations, would test whether the dependence on these two groups alone is complete.

Supplementary data. Supplementary material is available at https://doi.org/ ....

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